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2026 NCE Specimen Examination

2026

MATHEMATICS

Paper 4 of 4 | Inequalities · Transformations · Cumulative Frequency ·

Histograms

Time Allowed: 2 hours

Total Marks: 100

Candidates: Grade 9 NCE

Index Number: _____

Formulae

Area: Rectangle $A = lw$ | Triangle $A = \frac{1}{2}bh$

Volume: Cuboid $V = lwh$ | Cylinder $V = \pi r^2 h$

Statistics: Frequency density = frequency $\div$ class width

Transformations: Enlargement: image distance = $k \times$ object distance from centre

Circle $A = \pi r^2$ | Trapezium $A = \frac{1}{2}(a+b)h$

Cone $V = \frac{1}{3}\pi r^2 h$ | Sphere $V = \frac{4}{3}\pi r^3$

Estimated mean = $\frac{\sum fx}{\sum f}$ | Median at $\frac{1}{2}n$ on CF graph

Rotation 90° CW: $(x, y) \rightarrow (y, -x)$ | 90° CCW: $(x, y) \rightarrow (-y, x)$

Instructions to Candidates

1. Write your Index Number in the space provided above.
2. Answer ALL questions in Sections A, B and C.
3. In Section A, tick ONE box only for each question.
4. In Sections B and C, show all working clearly in the spaces provided.
5. Marks available for each question are shown in brackets [].
6. You may use a calculator. $\pi = 3.142$ unless otherwise stated.

7. This paper contains 28 questions totalling 100 marks.

SECTION A — Multiple Choice Questions (10 marks) [Q1–10, 1 mark each]

For each question, tick ONE box only.

- 1** Solve $2x + 1 > 7$. [1]
- ☐ A) $x > 3$
- ☐ B) $x > 4$
- ☐ C) $x > 6$
- ☐ D) $x < 3$
-
- 2** The image of point $(3, -2)$ after reflection in the y -axis is: [1]
- ☐ A) $(-3, -2)$
- ☐ B) $(3, 2)$
- ☐ C) $(-3, 2)$
- ☐ D) $(2, -3)$
-
- 3** A cumulative frequency graph is also known as: [1]
- ☐ A) A histogram
- ☐ B) An ogive
- ☐ C) A frequency polygon
- ☐ D) A pie chart
-
- 4** In a histogram, frequency density = [1]
- ☐ A) frequency $\times$ class width
- ☐ B) frequency $\div$ class width
- ☐ C) class width $\div$ frequency
- ☐ D) frequency + class width
-
- 5** On a number line, the inequality $x \geq 3$ is represented by: [1]
- ☐ A) A closed circle at 3 with an arrow pointing right
- ☐ B) An open circle at 3 with an arrow pointing right
- ☐ C) A closed circle at 3 with an arrow pointing left
- ☐ D) An open circle at 3 with an arrow pointing left
-
- 6** A translation by vector $\begin{pmatrix} -2 \\ 5 \end{pmatrix}$ moves point $(4, 1)$ to: [1]
- ☐ A) $(2, 6)$

- ☐ B) (6, -4)
- ☐ C) (-2, 5)
- ☐ D) (4, -1)

7 From a cumulative frequency graph for 60 values, the median is read at cumulative frequency: [1]

- ☐ A) 30
- ☐ B) 20
- ☐ C) 45
- ☐ D) 60

8 A class interval 10–20 has a frequency density of 3.5. The frequency is: [1]

- ☐ A) 3.5
- ☐ B) 7
- ☐ C) 35
- ☐ D) 350

9 The solution set of $x > -2$ AND $x \leq 4$ is: [1]

- ☐ A) $-2 < x \leq 4$
- ☐ B) $x > -2$ only
- ☐ C) $x \leq 4$ only
- ☐ D) $x < -2$ or $x > 4$

10 A regular hexagon has rotational symmetry of order: [1]

- ☐ A) 3
- ☐ B) 4
- ☐ C) 6
- ☐ D) 12

SECTION B — Short Structured Questions (48 marks) [Q11–22]

Show all working. Answers without working may not receive full marks.

11 Inequalities — Basic

[3]

- (a) Solve $3x - 5 \geq 7$.

[1]

- (b) Solve $2(x + 3) < 14$.

[1]

- (c) List all integer values of n such that $-3 < n \leq 2$.

[1]

12 Transformations — Reflections

[3]

- (a) Write down the coordinates of $A(2, 5)$ after reflection in the x -axis. [1]

- (b) Write down the coordinates of $B(-3, 4)$ after reflection in the y -axis. [1]

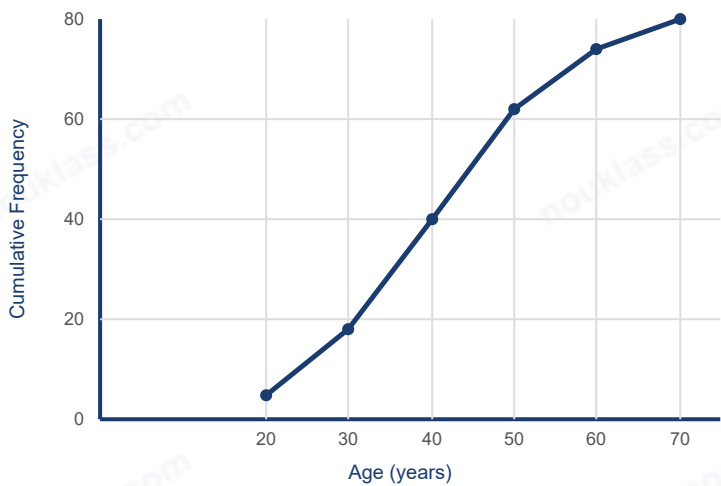
- (c) Write down the coordinates of $C(4, -2)$ after reflection in the line $y = x$. [1]

13

Cumulative Frequency — Reading a Graph

[4]

The cumulative frequency graph below shows the ages of 80 people at a community centre.



- (a) Use the graph to estimate the median age. Show your method on the graph. [2]

- (b) Estimate the interquartile range of the ages. Show your method on the graph. [2]

14

Histograms — Frequency Density

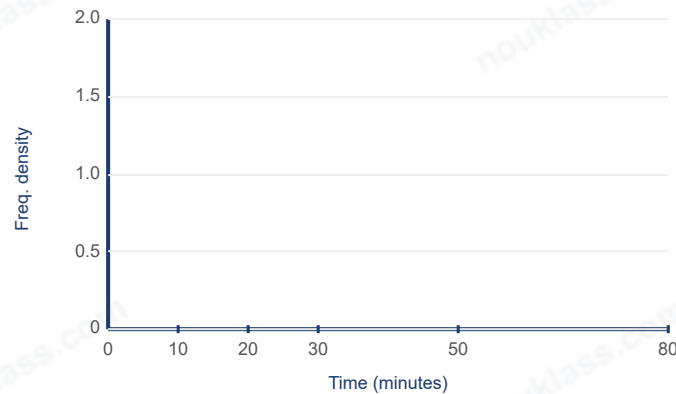
[4]

The table shows the times taken by 51 students to complete a task.

Time (min)	0–10	10–20	20–30	30–50	50–80
Frequency	8	15	12	10	6
Freq. density					

- (a) Complete the frequency density row in the table above. [2]

- (b) Using the axes below, draw the histogram for this data. [2]



15 Transformations — Translation and Rotation

[4]

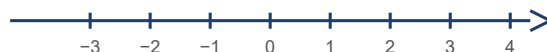
- (a) Triangle P has vertices at (1, 2), (3, 2) and (2, 4). It is translated by vector $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$. Write down the coordinates of the vertices of its image P'. [2]

- (b) Point Q(5, 3) is rotated 90° clockwise about the origin. Find the coordinates of its image Q'. [2]

16 Inequalities — Number Line and Solving

[4]

- (a) On the number line below, represent the inequality $-1 \leq x < 3$. [2]



- (b) Write the inequality: *x is at least 5 but less than 12*. [1]

- (c) Solve $5 - 2x > 1$. State clearly the direction of the inequality in your answer. [1]

17 Cumulative Frequency — Constructing a Table and Graph

[4]

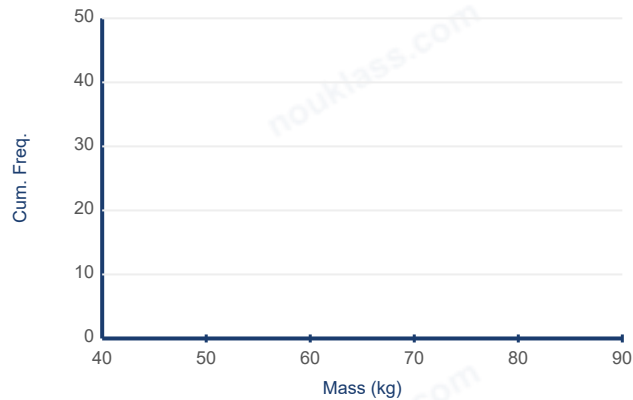
The masses of 50 students are given in the table below.

Mass (kg)	40–50	50–60	60–70	70–80	80–90
Frequency	4	12	18	11	5
Cumulative freq.					

- (a) Complete the cumulative frequency row in the table above. [2]

- (b) Plot the cumulative frequency graph on the grid below, using the upper class boundaries as your x values. Join the points with a

[2]



smooth curve.

18 Transformations — Enlargement

[5]

Rectangle ABCD has vertices A(1, 1), B(3, 1), C(3, 2) and D(1, 2). It is enlarged by scale factor 3 with the origin as the centre of enlargement.

- (a) Find the coordinates of the image A'B'C'D'. [3]

- (b) Find the ratio of the area of A'B'C'D' to the area of ABCD. [2]

19 Histograms — Interpreting

[4]

A histogram shows the distribution of test marks for 150 students.
The frequency densities (class width = 20 throughout) are:

- 0–20: frequency density = 0.5
- 20–40: frequency density = 2.0
- 40–60: frequency density = 3.0
- 60–80: frequency density = 1.5
- 80–100: frequency density = 0.5

(a) Calculate the frequency for each class and verify that the total is 150.

[2]

(b) Write down the modal class.

[1]

(c) Calculate the percentage of students who scored 60 marks or above.

[1]

20 Inequalities — Combined

[4]

Find all integer values of x that satisfy *both* of the following inequalities simultaneously:

$$2x + 3 > 7 \quad \text{and} \quad x \leq 6$$

Show your working clearly.

21 Transformations — Combined

[5]

Consider point $A(3, 1)$.

- (a) Point $A(3, 1)$ is rotated 90° anticlockwise about the origin to give A' . Write down the coordinates of A' . [2]

- (b) Point $A'(-1, 3)$ is reflected in the line $y = x$ to give A'' . Write down the coordinates of A'' . [2]

- (c) Describe fully the single transformation that maps $A(3, 1)$ directly to $A''(3, -1)$. [1]

22 Cumulative Frequency — Statistical Analysis

[4]

From a cumulative frequency graph for 100 students' test scores, the following values are estimated:

Lower quartile = 42, Median = 58, Upper quartile = 71

- (a) Calculate the interquartile range. [1]

- (b) A student is chosen at random. Estimate the probability that the student scored more than 71. [1]

- (c) Estimate the probability that a randomly chosen student scored between 42 and 71 (inclusive). [1]

- (d) Comment on the skewness of the distribution. Give a reason based on the quartile values. [1]

SECTION C — Long Structured Questions (42 marks) [Q23–28, 7 marks each]

Show ALL working clearly. Marks may be awarded for correct method even if the final answer is wrong.

23 Inequalities — Extended

[7]

- (a) Solve the inequality $3x - 7 < 2x + 5$.

[2]

- (b) Show your answer to (a) on the number line below.

[1]



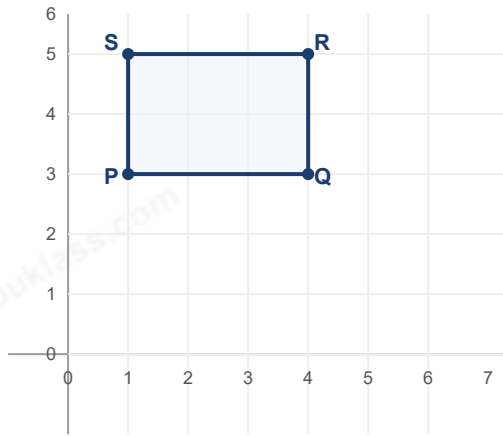
- (c) Find all integer values of x that satisfy *both* conditions simultaneously:
 $-1 \leq 2x - 3 < 7$ and $x < 5$.

[4]

24 Transformations — Coordinate Geometry

[7]

A square PQRS has vertices P(1, 3), Q(4, 3), R(4, 6) and S(1, 6).



- (a) Write down the name of quadrilateral PQRS and justify your answer by giving two properties. [1]

- (b) Reflect PQRS in the line $y = 3$ to give $P'Q'R'S'$. Write down the coordinates of P' , Q' , R' and S' . [3]

- (c) Rotate PQRS by 180° about the point $(1, 3)$. Write down the coordinates of the image. [3]

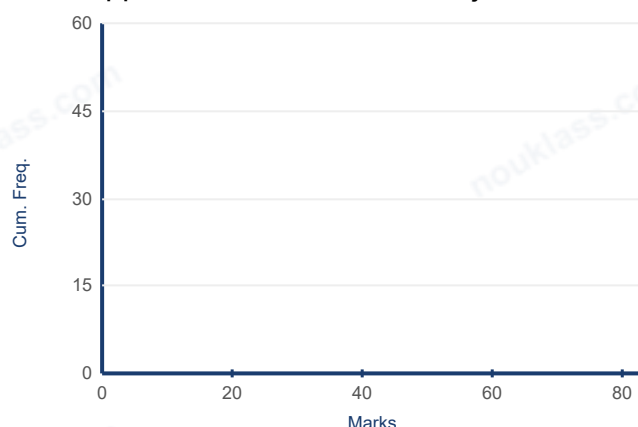
25 Cumulative Frequency — Full Analysis**[7]**

The marks (out of 80) of 60 students in a Science test are shown in the table below.

Marks	0–20	20–40	40–60	60–80
Frequency	5	18	27	10
Cumulative freq.				

(a) Complete the cumulative frequency row in the table above. [2]

(b) Draw the cumulative frequency graph on the axes below. Plot points at the upper class boundaries and join with a smooth curve. [2]



(c) Use your graph to estimate the median mark. [1]

(d) Estimate the interquartile range of the marks. Show clearly how you obtained your values. [2]

26 Histograms — Construction and Interpretation**[7]**

The table shows the times (in minutes) taken by 100 students to travel to school.

Time (min)	0–10	10–20	20–30	30–50	50–80
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Frequency	16	28	24	20	12
Freq. density					

- (a) Explain why a histogram, rather than a bar chart, should be used for this data. [1]

- (b) Complete the frequency density row in the table. [2]

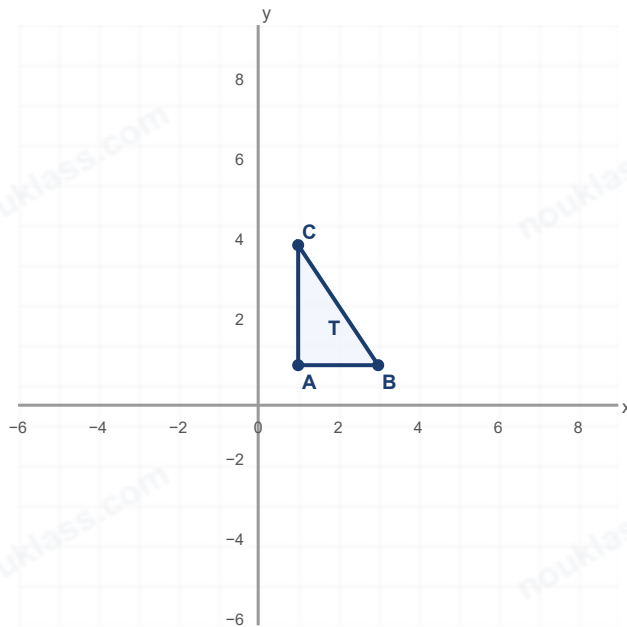
- (c) Verify that the total frequency is 100. Show your working. [1]

- (d) Calculate an estimate of the mean travel time. Use midpoints 5, 15, 25, 40 and 65. [3]

27 Transformations — Combined and Description

[7]

Triangle T has vertices at A(1, 1), B(3, 1) and C(1, 4).
A coordinate grid is shown below (x: -6 to 8, y: -6 to 8).



- (a) Rotate triangle T by 90° clockwise about the origin to give triangle T'. Write down the coordinates of the vertices of T'. [2]

- (b) Reflect triangle T in the line $y = -1$ to give triangle T''. Write down the coordinates of the vertices of T''. [2]

- (c) Describe fully the single transformation that maps triangle T directly onto triangle T''. [3]

28 Histograms and Cumulative Frequency — Combined

[7]

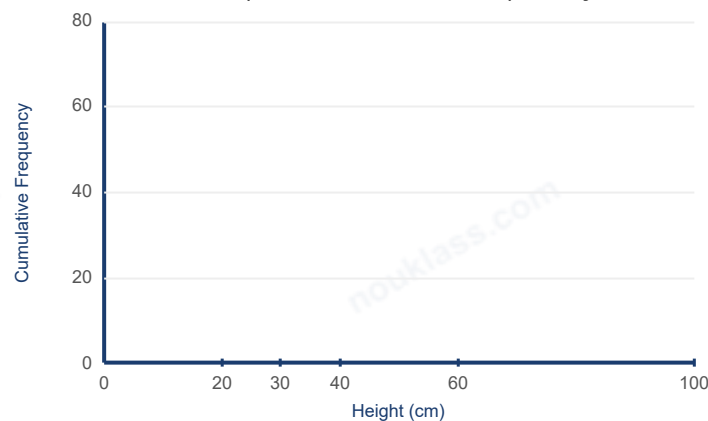
The heights (in cm) of 80 plants are recorded in the table below.

Height (cm)	0–20	20–30	30–40	40–60	60–100
Frequency	8	16	24	20	12
Freq. density					
Cumulative freq.					

- (a) Complete the frequency density row. Class widths are 20, 10, 10, 20 and 40. [2]

- (b) Complete the cumulative frequency row. [1]

- (c) On the axes below, draw the cumulative frequency graph. Plot points at upper class boundaries (20, 30, 40, 60, 100) and join with a smooth S-curve. [2]



S-curve.

- (d) Use your cumulative frequency graph to estimate the median height. [1]

- (e) Comment on the shape (skewness) of the distribution of plant heights. Give a reason. [1]

Mark Scheme — Paper 4 (Inequalities · Transformations · Cumulative Frequency · Histograms) 2026

This mark scheme is provided for teacher and self-study use. Equivalent correct methods receive full credit. For multi-step questions, method marks (M) may be awarded even if the final answer is wrong, provided the method is clearly shown.

Q	Answer / Working	Marks	Notes
SECTION A			
1	A — $x > 3$ ($2x > 6$, $x > 3$)	1	
2	A — $(-3, -2)$ (x-coordinate negated; y unchanged)	1	
3	B — An ogive	1	
4	B — frequency $\div$ class width	1	
5	A — Closed circle at 3, arrow pointing right	1	$x \geq 3$ uses closed circle
6	A — $(2, 6)$ ($4 + (-2) = 2$; $1 + 5 = 6$)	1	
7	A — Cumulative frequency 30 (median at $n/2 = 30$)	1	
8	C — 35 ($3.5 \times 10 = 35$)	1	
9	A — $-2 < x \leq 4$	1	
10	C — 6 (regular hexagon has 6-fold rotational symmetry)	1	
SECTION B			
11(a)	$3x \geq 12$; $x \geq 4$	1	
11(b)	$2x < 8$; $x < 4$	1	
11(c)	-2, -1, 0, 1, 2	1	-3 excluded (strict ineq.); 2 included
12(a)	(2, -5) (y-coordinate negated)	1	
12(b)	(3, 4) (x-coordinate negated)	1	
12(c)	(-2, 4) (coordinates swapped)	1	
13(a)	At CF = 40, read $x \approx$ 40 years from graph	2	M1 horizontal line at CF=40; A1 correct reading ± 1
13(b)	LQ at CF=20 $\rightarrow x \approx 33$; UQ at CF=60 $\rightarrow x \approx 48$; IQR $\approx$ 15 years	2	M1 both quartile readings; A1 IQR ± 2

14(a)	Freq densities: $8/10=0.8$; $15/10=1.5$; $12/10=1.2$; $10/20=0.5$; $6/30=0.2$	2	1 mark for any 3 correct; 2 for all 5
14(b)	Bars drawn at correct heights at x intervals 0–10, 10–20, 20–30, 30–50, 50–80	2	M1 correct widths; A1 correct heights
15(a)	$P' = (4, 1), (6, 1), (5, 3)$ (each vertex + $(3, -1)$)	2	1 mark for any 2 correct
15(b)	90° CW: $(x, y) \rightarrow (y, -x)$; $Q(5,3) \rightarrow Q'(3, -5)$	2	M1 correct rule; A1
16(a)	Closed circle at -1 , open circle at 3 , line segment between them	2	1 each endpoint symbol correct
16(b)	$5 \leq x < 12$	1	
16(c)	$5 - 2x > 1 \rightarrow -2x > -4 \rightarrow x < 2$ (inequality reverses when dividing by negative)	1	Must state direction of ineq.
17(a)	Cumulative frequencies: 4, 16, 34, 45, 50	2	1 mark for any 3 correct
17(b)	Points plotted at $(50,4), (60,16), (70,34), (80,45), (90,50)$; smooth S-curve drawn	2	M1 correct points; A1 smooth curve
18(a)	Multiply each coordinate by 3: $A'(3,3), B'(9,3), C'(9,6), D'(3,6)$	3	1 mark per pair of coordinates
18(b)	Area scale factor = $3^2 = 9$; Ratio = 9 : 1	2	M1 squaring scale factor; A1
19(a)	Freq: $0.5 \times 20 = 10$; $2.0 \times 20 = 40$; $3.0 \times 20 = 60$; $1.5 \times 20 = 30$; $0.5 \times 20 = 10$; Total = 150 ✓	2	M1 method; A1 all five + total
19(b)	40–60 (highest frequency density = 3.0)	1	
19(c)	$(30+10)/150 \times 100 = 40/150 \times 100 \approx$ 26.7%	1	Accept 26–27%
20	$2x + 3 > 7 \rightarrow x > 2$; combined with $x \leq 6$: $2 < x \leq 6$; integer values: 3, 4, 5, 6	4	M1 each inequality; M1 combining; A1 all four values
21(a)	90° CCW: $(x, y) \rightarrow (-y, x)$; $A(3,1) \rightarrow$ $A'(-1, 3)$	2	M1 rule; A1
21(b)	Reflection in $y=x$: $(x,y) \rightarrow (y,x)$; $A'(-1,3) \rightarrow$ $A''(3, -1)$	2	M1 rule; A1
21(c)	$A(3,1) \rightarrow A''(3,-1)$: Reflection in the x-axis	1	
22(a)	$IQR = 71 - 42 =$ 29	1	
22(b)	$P(\text{score} > 71) = 25/100 =$ $1/4$ (top quarter above UQ)	1	
22(c)	$P(42 \leq \text{score} \leq 71) = 50/100 =$ $1/2$ (middle half between quartiles)	1	

22(d)	UQ – Median = 13; Median – LQ = 16; Upper half shorter → distribution is negatively skewed (left skew)	1	Any sensible comment with reason
SECTION C			
23(a)	$3x - 7 < 2x + 5$; $x < 12$	2	M1 collecting terms; A1
23(b)	Open circle at 12 on number line, arrow pointing left	1	
23(c)	$-1 \leq 2x - 3 < 7 \rightarrow 2 \leq 2x < 10 \rightarrow 1 \leq x < 5$; Combined with $x < 5$: $1 \leq x < 5$; Integers: 1, 2, 3, 4	4	M1 solving compound ineq; M1 combining; A1 range; A1 integers
24(a)	PQRS is a square ; all sides = 3 units; all angles = 90°	1	
24(b)	Reflection in $y=3$: $P(1,3) \rightarrow P'(1,3)$; $Q(4,3) \rightarrow Q'(4,3)$; $R(4,6) \rightarrow R'(4,0)$; $S(1,6) \rightarrow S'(1,0)$	3	1 mark per pair
24(c)	180° rotation about (1,3): $P(1,3) \rightarrow P''(1,3)$; $Q(4,3) \rightarrow Q''(-2,3)$; $R(4,6) \rightarrow R''(-2,0)$; $S(1,6) \rightarrow S''(1,0)$	3	1 mark per pair; method: ($2 \times cx - x$, $2 \times cy - y$)
25(a)	CF: 5, 23, 50, 60	2	1 for any 2 correct
25(b)	Points plotted at (20,5),(40,23),(60,50), (80,60); smooth S-curve through points	2	M1 correct points; A1 smooth curve
25(c)	At CF = 30, mark $\approx$ 50 (from graph)	1	Accept 48–52
25(d)	LQ at CF=15 $\rightarrow \approx 38$; UQ at CF=45 $\rightarrow \approx 57$; IQR $\approx$ 19 marks	2	M1 both quartile readings; A1 IQR ± 3
26(a)	Unequal class widths — a bar chart assumes equal widths; histogram uses area (freq density) to represent frequency	1	
26(b)	$16/10=1.6$; $28/10=2.8$; $24/10=2.4$; $20/20=1.0$; $12/30=0.4$	2	1 for any 3 correct
26(c)	$1.6 \times 10 + 2.8 \times 10 + 2.4 \times 10 + 1.0 \times 20 + 0.4 \times 30 = 16+28+24+20+12 = \mathbf{100 \checkmark}$	1	
26(d)	$f \times x$: $16 \times 5=80$; $28 \times 15=420$; $24 \times 25=600$; $20 \times 40=800$; $12 \times 65=780$; $\Sigma fx=2680$; mean $= 2680/100 = \mathbf{26.8 \text{ min}}$	3	M1 midpoints; M1 Σfx ; A1 mean
27(a)	90° CW: $(x,y) \rightarrow (y,-x)$; $A(1,1) \rightarrow (1,-1)$; $B(3,1) \rightarrow (1,-3)$; $C(1,4) \rightarrow (4,-1)$	2	M1 rule; A1 all three
27(b)	Reflect in $y=-1$: $(x,y) \rightarrow (x,-2-y)$; $A(1,1) \rightarrow (1,-3)$; $B(3,1) \rightarrow (3,-3)$; $C(1,4) \rightarrow (1,-6)$	2	M1 correct rule; A1 all three

27(c)	Reflection in the line $y = -1$ (T maps directly to T" by this single reflection)	3	M1 type; M1 correct line; A1 full description
28(a)	8/20= 0.4 ; 16/10= 1.6 ; 24/10= 2.4 ; 20/20= 1.0 ; 12/40= 0.3	2	1 for 3 correct; 2 for all 5
28(b)	8, 24, 48, 68, 80	1	
28(c)	Points at (20,8),(30,24),(40,48),(60,68), (100,80); smooth S-curve	2	M1 correct points; A1 curve
28(d)	At CF = 40, read height $\approx$ 37 cm	1	Accept 35–39
28(e)	Positively skewed — most plants are shorter (data bunched at lower end, tail extends toward higher heights)	1	Any valid comment with reason
TOTAL		100	